

LinguaQuest: Integrating Linguistic and Cultural Learning in an Educational Role-Playing Game

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Abstract

Educational games offer engaging environments for learning, yet most systems address either linguistic instruction or cultural education in isolation. This separation limits our understanding of how language learning unfolds when linguistic structure and cultural context are integrated within a single interactive experience. We present an educational role-playing game that combines linguistic instruction with uniquely Canadian cultural content, embedding culturally situated vocabulary and scenarios directly into grammar- and syntax-focused game-play tasks. We evaluated the game using a pre- and post-quiz, alongside a player experience (PXI) survey. Results show substantial gains in both linguistic and cultural knowledge following a single game-play session. PXI responses further indicate strong ratings for enjoyment, curiosity, and immersion, suggesting that the integrated design supports learning while maintaining a positive game-play experience. These preliminary findings indicate that combining linguistic and cultural learning within a unified game-based environment has a potential to be both educationally effective and experientially engaging.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; *User studies*; • **Applied computing** → *Computer-assisted instruction*; • **Computing methodologies** → *Game design*.

Keywords

educational games, turn-based RPG, linguistics education, cultural learning, user study, player experience, serious games

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1 Introduction

Educational games offer interactive and engaging environments for learning, yet it remains unclear how effectively they can support domains that intertwine linguistic concepts with cultural knowledge. In this work, we investigate whether a 2D role-playing game (RPG) can facilitate learning across both areas within a single interactive experience.

Prior research in digital game-based language learning demonstrates that embedding linguistic tasks within gameplay and narrative can support vocabulary acquisition, grammatical awareness, and contextualized language use [6]. Similarly, cultural education games show that narrative and role-play enable learners to engage with culturally situated meaning through interaction rather than explicit instruction [9]. Standardized instruments such as the Player Experience Inventory (PXI) provide validated measures of engagement, competence, autonomy, and affect during gameplay [1]. Together, these strands motivate an integrated examination of learning and player experience in educational games.

However, most educational games address linguistic or cultural learning in isolation, leaving limited empirical evidence on how learners engage when both are presented together in a unified interactive setting.

To address this gap, we developed LingaQuest – a lightweight 2D educational RPG that integrates linguistic instruction with Canadian cultural content. Cultural scenarios and vocabulary are embedded directly into gameplay mechanics that support linguistic reasoning, enabling players to engage with both domains through exploration and problem solving. We evaluated the game using a pre-test/post-test design alongside a standardized PXI survey. Results indicate measurable improvements in linguistic and cultural knowledge, alongside strong engagement-related PXI ratings, suggesting that the integrated design supports learning while maintaining a positive player experience.

This paper makes the following contributions:

- A 2D educational turn-based RPG that integrates linguistic learning with culturally grounded content.



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- A preliminary user study that shows learning gains in linguistic and cultural knowledge along with a player experience evaluation.

2 Related Work

Experiential learning is gaining traction as an alternative pedagogy, especially through "serious games." These games integrate educational content into interactive environments for fields like health, management, and disaster prevention, see e.g., [3][8][10][12]. A specific example, RPGs in both tabletop and digital formats, immerse students in simulated scenarios. By requiring them to adopt another's persona and viewpoints, this "embodied" learning offers a promising pathway for developing intercultural competence.

PyraMaze VR [11] significantly improved learners' comprehension of ancient Egyptian culture, achieving a 34.7% knowledge gain compared to only 13.17% through textbook learning and received high satisfaction ratings, indicating both cognitive and affective benefits. Similarly, The Travel Coser [2] markedly increased participants' understanding of storytelling and West Lake culture, with statistically significant improvements in cultural learning. In more immersive contexts, the Stardew Valley Chatbot [7] study showed that conversational interaction with the AI character "Daisy" deepened players' sense of presence and continuity within the game world, reinforcing RPGs' potential for narrative-based engagement. ReWIND [5] effectively visualized emotional states through adaptive color saturation tied to players' progress, supporting emotion regulation and self-awareness. Finally, TogetherTales RPG [4] successfully promoted prosocial behaviour in children through cooperative role-play enhanced by AI and AR technologies. Collectively, these findings highlight that RPG-based systems can deliver meaningful, measurable, and emotionally resonant learning experiences, making them powerful tools for education, therapy, and cultural exploration. They can be highly effective in enhancing learning outcomes, emotional engagement, and cultural understanding across diverse domains.

3 LinguaQuest: Process, Design, and Implementation

The overall game play sequence is illustrated in Figure 1 showing the various scenes and the mini-games.

The player is first prompted to choose the world they would like to play. This will be extended later to include additional worlds. The game begins with the player character entering *Canada World*. The initial scene is set indoors, within a room containing interactive objects and progression checkpoints. A guiding fox provides instructions and contextual guidance throughout this scene. Various culturally representative objects are scattered throughout the room, such as hockey equipment and poutine. Interacting with these objects introduces players to their cultural significance. To complete the level and exit the room, the player must complete a short trivia challenge based on the learned content.

In the second scene, the player transitions to an outdoor environment. Collectible items such as hearts (representing lives), coins, and treasure chests are placed throughout the area. Opening a treasure chest reveals a linguistic learning tree diagram, where acquired

learning elements are visually stored and tracked. Additional mechanics include patches of black ice that cause the player to slide, reinforcing Canadian environmental themes through gameplay.

As progression continues, the player opens a second treasure chest and obtains a magic lasso, which becomes a core mechanic in the subsequent mini-game. Shortly after, the player encounters their first enemy, triggering the second mini-game. The second mini-game challenges players to reconstruct out-of-order sentences. The sentences test both linguistic understanding and cultural knowledge, as they are thematically tied to Canadian contexts. Each correct answer inflicts damage on the enemy, and the player succeeds by correctly reconstructing all sentences.

The third scene is set during the autumn season in Canada. Opening a treasure chest reveals a book, after which the player encounters a squirrel that requests to be fed. This interaction initiates the third mini-game. In this mini-game, the player uses the magic lasso to connect sentence fragments in the correct order. The task continues to reinforce both linguistic structure and culturally themed content. Later in the third scene, the player encounters a moose. Interacting with the moose initiates the fourth mini-game, which focuses on word-level discrimination, where the player selects between red and blue shockwaves to target word blocks. Correct word blocks are eliminated using blue shockwaves, while incorrect ones are targeted with red. This concludes the first world of the game.

The cultural elements included in LinguaQuest were selected to reflect vocabulary and concepts that are distinctively Canadian and relevant to everyday life. Items were chosen to balance historical, linguistic, and regional aspects of Canadian culture, such as traditional clothing (e.g., parka, moccasin), recreational activities (hockey, portage), seasonal phenomena (black ice, snowshoe), and iconic foods (maple syrup, poutine). While no formal curriculum or standardized framework was used, selections were guided by widely recognized Canadian cultural references and intended to provide learners with meaningful, contextually grounded exposure. This approach emphasizes authentic cultural knowledge over stereotypical or tokenistic representations and aims to integrate cultural learning seamlessly with linguistic instruction.

4 Methodology

4.1 Study Design

The study used a within-subjects design comparing performance before and after, with participants completing knowledge tests and a player-experience survey around a single gameplay session. The study was conducted primarily online via an itch.io WebGL deployment, with seven participants completing the same procedure remotely. One participant completed the study in person using identical materials and instructions. This structure allowed us to measure changes in linguistic and cultural knowledge and to capture subjective gameplay experience. All participants completed the same sequence of tasks to ensure consistency across sessions. This sequence allowed us to isolate the effect of the gameplay session on both learning outcomes and reported experience, which are central to evaluating whether the game supports linguistic and cultural learning in a unified way. This study was reviewed and approved by the Research Ethics Board at Ontario Tech University (REB #19510). All participants provided informed consent prior to participation.



Figure 1: Game scenes from LinguaQuest. Eight representative scenes illustrating environments, interactions, and cultural elements in the Canadian world.

Table 1: Participant background familiarity ratings (1 = Not familiar at all, 5 = Very familiar).

Participant	Video Games	Linguistics	Canadian Culture
P1	5	4	4
P2	4	3	5
P3	4	4	4
P4	3	3	3
P5	2	3	2
P6	5	5	4
P7	4	4	1
P8	1	1	1
Mean	3.50	3.38	3.00
Median	4.00	3.50	3.50
SD	1.41	1.19	1.51

4.2 Participants

Eight participants ($N = 8$) were recruited through convenience sampling and mostly included undergraduate computer science students. Ages varied, with most individuals in their early to mid-twenties. In addition to basic age information, we systematically collected background variables relevant to learning outcomes in a language-learning game context, including prior gaming experience, familiarity with Canadian culture, and prior linguistic knowledge.

Participants exhibited heterogeneous backgrounds across these dimensions. Self-reported familiarity ratings (1–5 Likert scale) indicated substantial variation in prior exposure to video games, linguistics, and Canadian culture. Familiarity with Canadian culture ranged from very low to very high across participants, while linguistic familiarity clustered around mid-level ratings with notable individual differences. Gaming familiarity also varied, spanning low to high self-reported familiarity. These background differences are relevant given that prior familiarity with games, language-related concepts, and cultural context may influence both gameplay behavior and observed learning outcomes.

To improve interpretability, these background variables are summarized in a participant characteristics table (Table 1), reporting individual-level prior exposure to Canadian culture, linguistic background, and gaming experience. These variables provides important context for interpreting observed learning and engagement outcomes, and enables preliminary qualitative analysis of how prior knowledge may have shaped player experience.

No additional demographic variables were gathered due to the informal preliminary nature of the study, and participation required only basic computer and Internet access. Participants were not compensated.

4.3 Apparatus

4.3.1 Game Prototype. Participants played LingaQuest – our custom-built educational RPG developed in Unity (version: 2022.3.46f1).

Each session lasted approximately 10 to 15 minutes. The game introduced players to both linguistic concepts and uniquely Canadian cultural vocabulary through the sequence of gameplay (see Section 3).

4.3.2 Learning Assessments. Knowledge was assessed using a short quiz administered before and after gameplay. The knowledge quiz included ten short-answer items designed to test recognition and understanding of uniquely Canadian vocabulary. The full quiz is included in the Appendix. Each question was worth one point, with multi-step items awarding one point per correct component. Scores were summed separately for linguistic and cultural domains which allowed us to measure whether the game supported learning across the two areas it was designed to integrate.

4.4 Procedure

All participants followed the same procedure. At the start, they were informed about the study's goals and the sequence of tasks they would complete. After providing consent, participants began with a pre-test that measured both linguistic and cultural knowledge. The pre-test established a baseline so that any subsequent improvement could be attributed to the gameplay session.

Participants then played the game through the WebGL build hosted on the project's itch.io page. Instructions for access and controls were provided on the page, ensuring consistent onboarding. Gameplay typically lasted 10–15 minutes. During this phase, no feedback was given beyond what was already included in the game, allowing the session to reflect the game's intended standalone educational design.

Immediately after gameplay, participants completed the same combined linguistic and cultural post-test. Using the same instrument allowed direct comparison with the pre-test, enabling us to measure learning gains attributable to the session. Participants then completed the PXI [1] survey along with a brief set of background questions. Finally, participants were thanked for their time.

5 Results and Discussion

Participants showed substantial improvement from pre-test to post-test across both linguistic and cultural knowledge domains, as well as overall, following gameplay. Player experience responses further indicate that the prototype was generally perceived as engaging and enjoyable.

5.1 PXI

Overall, PXI item responses were skewed toward the positive end of the scale, suggesting that participants generally perceived the prototype as enjoyable and audio-visually appealing. See Figure 2. Across PXI constructs (−3 to +3), the highest mean ratings were for enjoyment and audiovisual appeal (2.58), followed by curiosity (2.54), autonomy (2.17), and immersion (1.96), whereas goal clarity and progress feedback were mid-range, and perceived competence and control were lowest (ease of control = 1.29; mastery = 1.08).

Taken together, these findings indicate that the prototype performs strongly on affective and engagement-related dimensions (e.g., enjoyment, curiosity, and audiovisual appeal). However, there

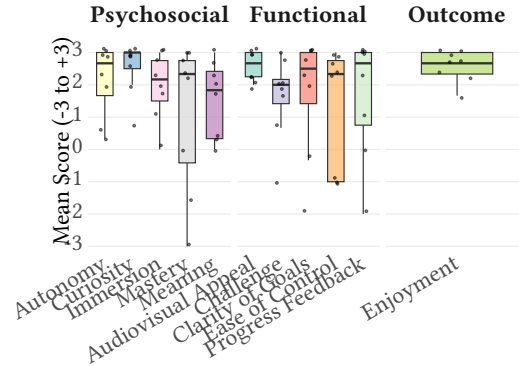


Figure 2: PXI construct score distributions. Distribution of PXI construct scores (scale −3 to +3), showing highest ratings for enjoyment, audiovisual appeal, and curiosity, with comparatively lower ratings for ease of control and mastery.

remains scope to strengthen interaction fluency and skill acquisition (e.g., control learnability, feedback interpretability, and perceived mastery). This pattern is consistent with the project's design objective of delivering an engaging exploration-driven learning experience. It also implies that targeted usability improvements could further support learning by reducing extraneous cognitive load during mini-game interactions.

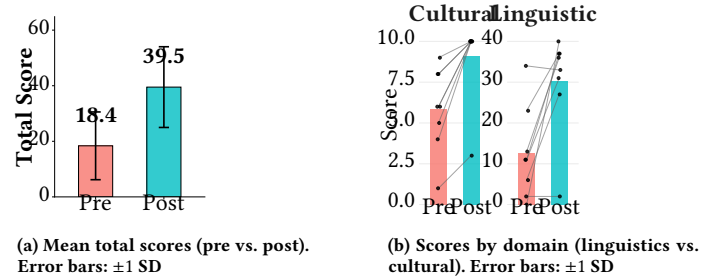


Figure 3: Pre/post learning outcomes. Aggregate and individual learning results before and after gameplay. Mean total scores increase substantially following gameplay; domain-level results show a ceiling effect in the cultural post-test;

5.2 Domain Knowledge

A paired-samples *t*-test revealed a statistically significant increase in total domain knowledge scores from pre-test ($M = 18.38$, $SD = 12.19$) to post-test ($M = 39.50$, $SD = 14.52$), $t(7) = 4.71$, $p < .005$, $d = 1.67$. This represents a 114.9% improvement relative to baseline performance.

Similarly, a paired-samples *t*-test showed a significant improvement in cultural knowledge scores from pre-test ($M = 5.88$, $SD = 2.59$) to post-test ($M = 9.12$, $SD = 2.47$), $t(7) = 5.25$, $p < .001$, $d = 1.86$, corresponding to a 55.1% increase relative to baseline.

Linguistic knowledge scores also increased significantly from pre-test ($M = 12.50$, $SD = 11.25$) to post-test ($M = 30.38$, $SD = 12.16$), $t(7) = 4.20$, $p < .005$, $d = 1.49$, reflecting a 143.0% improvement relative to baseline. For scores by domain, refer to Figure 3.

Domain-level interpretation. At the domain level, the cultural subscore exhibited a pronounced upward shift, with evidence of ceiling effects in the post-test distribution. This pattern suggests that many participants achieved near-maximal performance on the 10-item cultural quiz following gameplay. In contrast, the linguistics subscore accounted for the largest proportion of the overall gain, a finding that is consistent with the greater weighting of linguistics items (30 items) in the combined assessment.

Individual differences. Analysis of individual learning trajectories indicates that the majority of participants demonstrated meaningful improvement across both domains, although a small subset showed comparatively limited total gain. These results suggest that (i) the game may not be equally effective across all learner backgrounds and (ii) learning outcomes from a single, short gameplay session may vary as a function of prior knowledge, engagement, or interaction quality.

6 Limitations and Future work

This evaluation is constrained by (i) a small sample size, (ii) convenience sampling through personal networks, and (iii) reuse of the same instrument for pre- and post-tests, which may introduce practice effects. The study was conducted exclusively with Canadian students, which may limit generalizability. Nevertheless, we observed consistent learning improvements and expect similar effects to transfer to other cultural worlds we build. These limitations motivate follow-up studies with larger, more diverse samples and the use of counterbalanced or parallel quiz forms.

We are working on additional playable worlds that incorporate culturally grounded linguistic and aesthetic elements: Chinese and Japanese worlds. Each world will be designed with culturally representative environments, artifacts, and narrative cues, enabling comparative evaluation of how cultural contextualization influences engagement, motivation, and learning outcomes. This extension will also allow investigation of the generalizability of the observed effects beyond the Canadian context and support cross-cultural analysis of player experience and learning transfer.

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References

- [1] Vero Vanden Abeele, Katta Spiel, Lennart Nacke, Daniel Johnson, and Kathrin Gerling. 2020. Development and validation of the player experience inventory: A scale to measure player experiences at the level of functional and psychosocial consequences. *International Journal of Human-Computer Studies* 135 (2020), 102370. doi:10.1016/j.ijhcs.2019.102370
- [2] Defu Bao, Qinqing Fu, and Danni Shen. 2020. Gamification of Tourism Culture Learning: An Analysis of Immersive Games. In *2020 12th International Conference on Intelligent Human-Machine Systems and Cybernetics (IHMSC)*, Vol. 2. 36–39. doi:10.1109/IHMSC49165.2020.10087
- [3] Polona Caserman, Katrin Hoffmann, Philipp Müller, Marcel Schaub, Katharina Straßburg, Josef Wiemeyer, Regina Bruder, and Stefan Göbel. 2020. Quality criteria for serious games: serious part, game part, and balance. *JMIR serious games* 8, 3 (2020), e19037.
- [4] Riddhi Divanji, Aayushi Dangol, Ella J. Lombard, Katharine Chen, and Jennifer D. Rubin. 2024. TogetherTales RPG: Prosocial Skill Development Through Digitally Mediated Collaborative Role-Playing. In *Proceedings of the 23rd Annual ACM Interaction Design and Children Conference (Delft, Netherlands) (IDC '24)*. Association for Computing Machinery, New York, NY, USA, 1012–1015. doi:10.1145/3628516.3662048
- [5] Yew Ken Heng. 2021. ReWIND: Psychoeducation Game Leveraging Cognitive Behavioral Therapy (CBT) to Enhance Emotion Control for Generalized Anxiety Disorder. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI EA '21). Association for Computing Machinery, New York, NY, USA, Article 497, 5 pages. doi:10.1145/3411763.3451851
- [6] Hsiu-Ting Hung, Jie Chi Yang, Gwo-Jen Hwang, Hui-Chun Chu, and Chun-Chieh Wang. 2018. A scoping review of research on digital game-based language learning. *Computers & Education* 126 (2018), 89–104. doi:10.1016/j.compedu.2018.07.001
- [7] Jungmin Lee, Seoyoung Yoon, Hwajin Shim, and Youngjae Yoo. 2025. Development of an LLM-Based Chatbot to Support Learnability in Stardew Valley: A Diary Study Approach. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, Article 1070, 19 pages. doi:10.1145/3706598.3713310
- [8] Fernando Rodríguez López, Mario Arias-Oliva, Jorge Pelegrín-Borondo, and Luz María Marín-Vinuesa. 2021. Serious games in management education: An acceptance analysis. *The International Journal of Management Education* 19, 3 (2021), 100517.
- [9] Michela Mortara, Chiara Eva Catalano, Francesco Bellotti, Giusy Fiucci, Minica Houry-Panchetti, and Panagiotis Petridis. 2014. Learning cultural heritage by serious games. *Journal of Cultural Heritage* 15, 3 (2014), 318–325. doi:10.1016/j.culher.2013.04.004
- [10] Nahid Sharifzadeh, Hadi Kharrazi, Elham Nazari, Hamed Tabesh, Maryam Edalati Khodabandeh, Somayeh Heidari, and Mahmood Tara. 2020. Health education serious games targeting health care providers, patients, and public health users: scoping review. *JMIR serious games* 8, 1 (2020), e13459.
- [11] Intanon Tangtung, Luksamee Lunsaiy, and Sirion Vittayakorn. 2024. PyraMaze VR: An Immersive Gameplay for Ancient Egypt Learning. In *2024 16th International Conference on Information Technology and Electrical Engineering (ICITEE)*. 1–6. doi:10.1109/ICITEE62483.2024.10808623
- [12] Meng-Han Tsai, Yu-Lien Chang, Jen-Shiuan Shiau, and Shun-Mei Wang. 2020. Exploring the effects of a serious game-based learning package for disaster prevention education: The case of Battle of Flooding Protection. *International journal of disaster risk reduction* 43 (2020), 101393.